

# Laboratory Practical / Experiments Report

One of the requirements for obtaining an oral/practical degree for a course

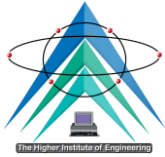
Networks (2), CCE 463  
Fourth Year  
Computers and Control Eng. (CCE) Program  
Regulation (2019)  
Second Semester, 2025/2026

- **Student name:** يكتب الاسم رباعي طبقا لما هو موجود في كشف التسجيل للطلاب باللغة الإنجليزية
- **Student Code:**
- **Section number:**
- **Student number in the registration statement:** يكتب رقم الطالب في كشف التسجيل في الفصل



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## Important Instructions:

- The student must complete the experiment at the end of the practical session and go to the instructor to evaluate the experiment.
- Experiments are carried out on the packet tracer and eNSP simulators.
- Students must be interested in writing, drawing and pictures.
- The practical / experiment report should be well coordinated.
- I hope all students will prepare this report independently and will not seek transfer assistance from their colleagues.
- If similar reports exist, these students will be evaluated with a zero score.
- The student can add any pages he wishes to add after the end of each experiment if he considers the general format of the report.
- Failure to comply with the above important instructions will expose you to losing scores.



**Student Name:**

**Student code:**

**Student number in the registration statement:**

| Experiment No.                                      | Evaluation                           |
|-----------------------------------------------------|--------------------------------------|
| Experiment (1): Review on IPs and Subnetting        | $\frac{\quad}{5}$                    |
| Experiment (2): Static Routing and Configuration    | $\frac{\quad}{5}$                    |
| Experiment (3): RIPv2 Routing and Configuration     | $\frac{\quad}{5}$                    |
| Experiment (4): EIGRP Routing and Configuration     | $\frac{\quad}{5}$                    |
| Experiment (5): OSPF Routing and Configuration      | $\frac{\quad}{5}$                    |
| Experiment (6): Telnet                              | $\frac{\quad}{5}$                    |
| Experiment (7): Wide Area Network (WAN)             | $\frac{\quad}{5}$                    |
| Experiment (8): Virtual Private Network (VPN)       | $\frac{\quad}{5}$                    |
| Experiment (9): Spanning Tree Protocol (STP)        | $\frac{\quad}{5}$                    |
| Experiment (10): Border Gateway Protocol (BGP)      | $\frac{\quad}{5}$                    |
| Experiment (11): Virtual Local Area Network (VLAN)  | $\frac{\quad}{5}$                    |
| Experiment (12): Network Design Using GNS3 Emulator | $\frac{\quad}{5}$                    |
| <b>Final Evaluation</b>                             | <b><math>\frac{\quad}{60}</math></b> |

*Signature*

**Dr. Mohamed A. Abdelhamed**  
**Eng.**



**Student Name:**

**Student code:**

**Student number in the registration statement:**

| Experiment No.                                      | Evaluation       |
|-----------------------------------------------------|------------------|
| Experiment (1): Review on IPs and Subnetting        | <u>5</u>         |
| Experiment (2): Static Routing and Configuration    | <u>5</u>         |
| Experiment (3): RIPv2 Routing and Configuration     | <u>5</u>         |
| Experiment (4): EIGRP Routing and Configuration     | <u>5</u>         |
| Experiment (5): OSPF Routing and Configuration      | <u>5</u>         |
| Experiment (6): Telnet                              | <u>5</u>         |
| Experiment (7): Wide Area Network (WAN)             | <u>5</u>         |
| Experiment (8): Virtual Private Network (VPN)       | <u>5</u>         |
| Experiment (9): Spanning Tree Protocol (STP)        | <u>5</u>         |
| Experiment (10): Border Gateway Protocol (BGP)      | <u>5</u>         |
| Experiment (11): Virtual Local Area Network (VLAN)  | <u>5</u>         |
| Experiment (12): Network Design Using GNS3 Emulator | <u>5</u>         |
| <b>Final Evaluation</b>                             | <b><u>60</u></b> |

*Signature*

**Dr. Mohamed A. Abdelhamed**  
**Eng.**

# **Experiment (1)**

## **Review on IPs and Subnetting**

### **1.1 Objectives**

### **1.2 Equipment / Program Requirements:**

### **1.3 Experiment Setup:**



## **1.4 Simulation Results:**

## **1.5 Conclusions:**

# **Experiment (2)**

## **Static Routing and Configuration**

### **2.1 Objectives**

### **2.2 Equipment / Program Requirements:**

### **2.3 Experiment Setup:**



## **2.4 Simulation Results:**

## **2.5 Conclusions:**

# **Experiment (3)**

## **RIPv2 Routing and Configuration**

### **3.1 Objectives**

### **3.2 Equipment / Program Requirements:**

### **3.3 Experiment Setup:**



### **3.4 Simulation Results:**

### **3.5 Conclusions:**

# **Experiment (4)**

## **EIGRP Routing and Configuration**

### **4.1 Objectives**

### **4.2 Equipment / Program Requirements:**

### **4.3 Experiment Setup:**



#### **4.4 Simulation Results:**

## **4.5 Conclusions:**

# **Experiment (5)**

## **OSPF Routing and Configuration**

### **5.1 Objectives**

### **5.2 Equipment / Program Requirements:**

### **5.3 Experiment Setup:**



## **5.4 Simulation Results:**

## **5.5 Conclusions:**

# **Experiment (6)**

## **Telnet**

### **6.1 Objectives**

### **6.2 Equipment / Program Requirements:**

### **6.3 Experiment Setup:**



## **6.4 Simulation Results:**

## **6.5 Conclusions:**

# **Experiment (7)**

## **Wide Area Network (WAN)**

### **7.1 Objectives**

### **7.2 Equipment / Program Requirements:**

### **7.3 Experiment Setup:**



## **7.4 Simulation Results:**

## **7.5 Conclusions:**

# **Experiment (8)**

## **Virtual Private Network (VPN)**

### **8.1 Objectives**

### **8.2 Equipment / Program Requirements:**

### **8.3 Experiment Setup:**



## **8.4 Simulation Results:**

## **8.5 Conclusions:**

# **Experiment (9)**

## **Spanning Tree Protocol (STP)**

### **9.1 Objectives**

### **9.2 Equipment / Program Requirements:**

### **9.3 Experiment Setup:**



## **9.4 Simulation Results:**

## **9.5 Conclusions:**

# **Experiment (10)**

## **Border Gateway Protocol (BGP)**

### **10.1 Objectives**

### **10.2 Equipment / Program Requirements:**

### **10.3 Experiment Setup:**



## **10.4 Simulation Results:**

## **10.5 Conclusions:**

# **Experiment (11)**

## **Virtual Local Area Network (VLAN)**

### **11.1 Objectives**

### **11.2 Equipment / Program Requirements:**

### **11.3 Experiment Setup:**



## **11.4 Simulation Results:**

## **11.5 Conclusions:**

# **Experiment (12)**

## **Network Design Using GNS3 Emulator**

### **12.1 Objectives**

### **12.2 Equipment / Program Requirements:**

### **12.3 Experiment Setup:**



## **12.4 Simulation Results:**

## **12.5 Conclusions:**

